

Reflections

By using the classic Iris dataset in the Module 02 lab exercises, I gained a better understanding of how machine learning libraries work together. From importing packages to grouping data and creating visualizations, these exercises taught me many useful things.

1. Dataset Structure and Pandas Aggregation

By checking the shape of the Iris dataset (150, 4) and grouping the data by species, I learned that understanding the data structure is important before model training. For example, the average petal length of *setosa* is 1.46 cm, while *virginica* is 5.55 cm. These numbers helped me see the differences between the species. Pandas also made it easier to group and calculate the data.

2. NumPy Calculations and Statistical Verification

Using NumPy's `np.mean` and `np.std` to calculate the mean and standard deviation of sepal length helped me better understand array vectorization. I also used assertion statements, such as `assert isinstance(...)`, to check the results. This taught me that checking intermediate results is just as important as doing the calculations.

3. Data Visualization

Using Matplotlib to create bar charts of species distribution helped me understand the data more clearly. The chart showed that each of the three classes had 50 samples, which means the dataset was balanced. Through this exercise, I learned that a balanced class distribution can reduce potential bias in machine learning and provide a good starting point for feature analysis.

